



科学研究的人工智能方法学

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未来25年，这125个问题将让科学家“大伤脑筋”

这125个问题，其中涉及生命科学的问题占46%，关系宇宙和地球的问题占16%，与物质科学相关的问题占14%以上，认知科学问题占9%。其余问题分别涉及数学与计算机科学、政治与经济、能源、环境和人口等。

慢思考时代下的大模型



Openai 的 o1 的综合评测

<https://arxiv.org/pdf/2409.18486>

Evaluation of OpenAI o1: Opportunities and Challenges of AGI

编程挑战

放射学报告生成

芯片设计任务

高中数学推理

人类学和地质学

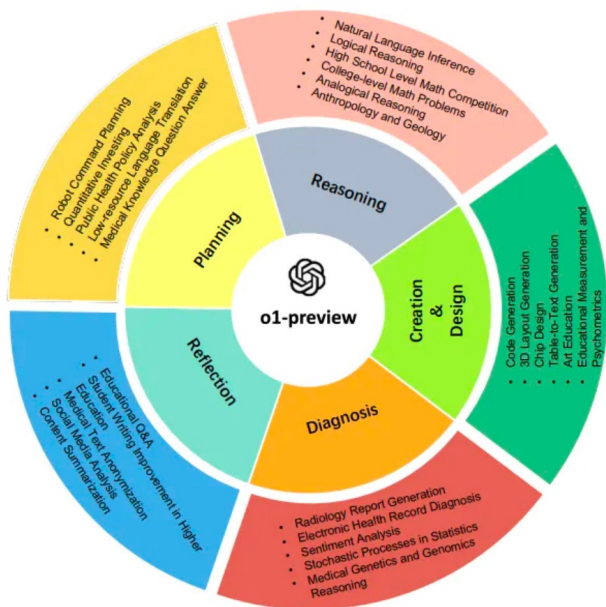
自然语言推理

社交媒体分析

量化投资

在AIME 2024数学竞赛中，o1模型的准确率达到惊人的83.3%，相比之下GPT-4o的准确率只有13.4%，提升6倍。

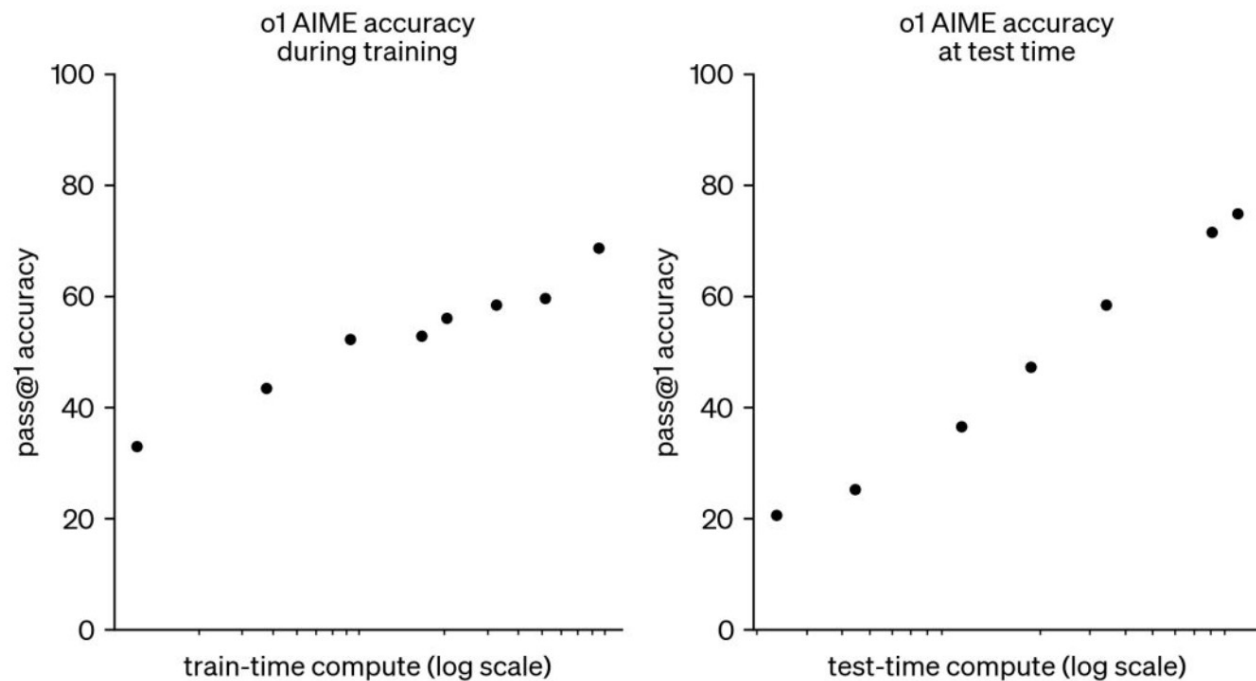
在CodeForces代码竞赛中，o1的准确率甚至达到了89%，GPT-4o的准确率是11.0%，呈现大幅提升。





Test/inference的scaling 的定义

大模型的性能会随着强化学习时间（训练时计算量）和推理时间（测试时计算量）的增加而显著提高



从 pretrain scaling law 要红利 转向 从 post training (test/inference scaling law) 要红利



Test/inference的scaling 如何获取

学会与大模型互动

优化推理输入：**prompt**

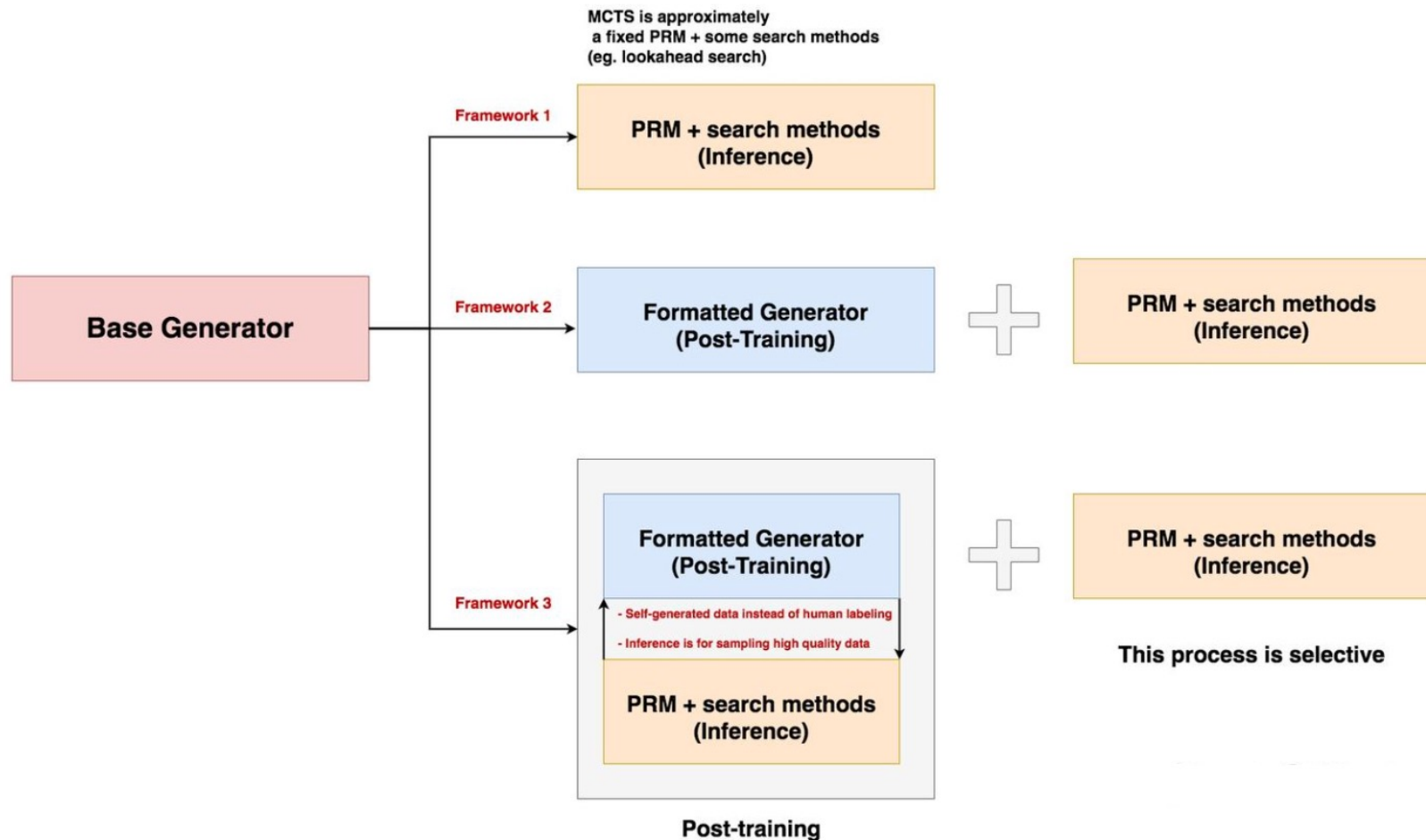
- Chain-of-Thought Prompting Elicits Reasoning in Large Language Models
 - Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, Denny Zhou

→ 优化推理输出：**revise output distribution**

让模型吃下一个问题后，自动化地去做CoT的过程



Test time scaling law 框架



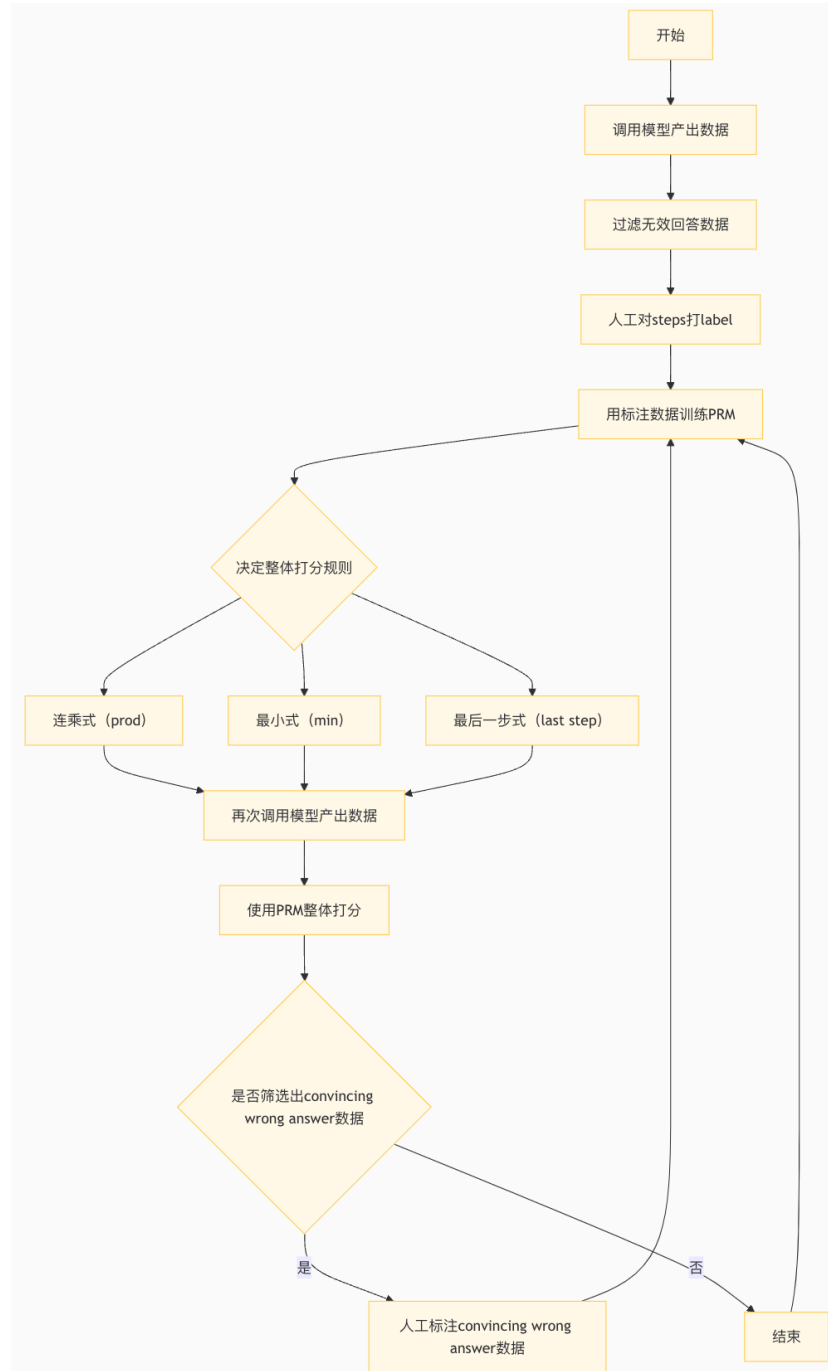


训练PRM

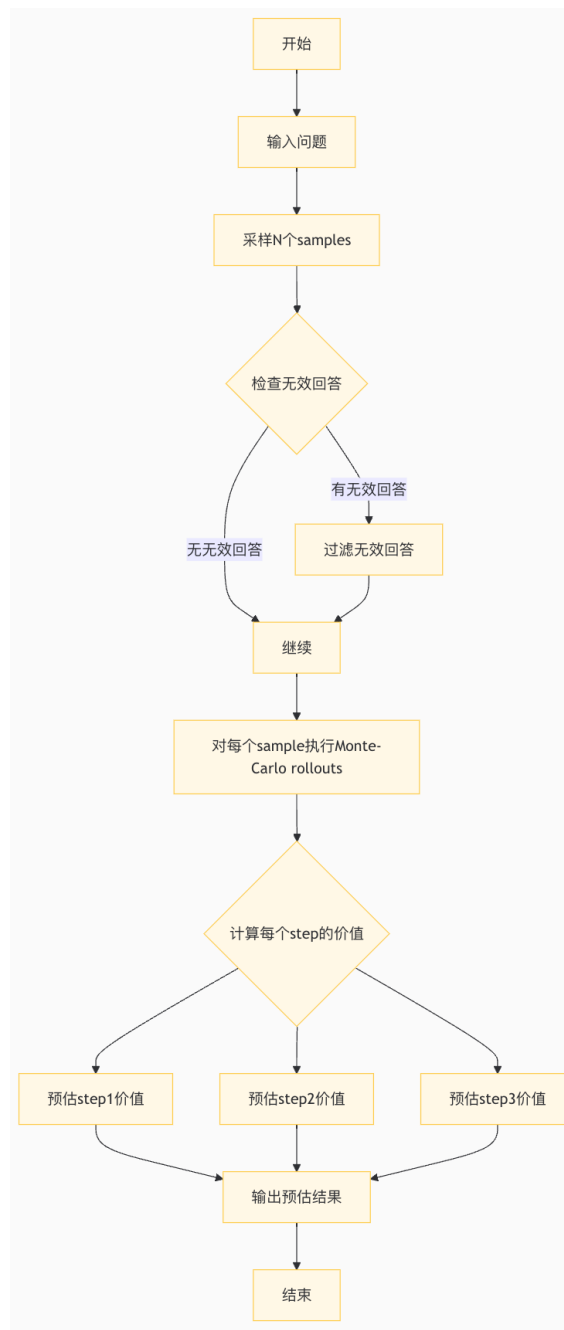
```

{
  "problem": "原始问题描述",
  "final_answer": "最终答案",
  "steps": [
    {
      "step": "步骤1的描述",
      "label": "positive/negative/neutral"
    },
    {
      "step": "步骤2的描述",
      "label": "positive/negative/neutral"
    },
    ...
  ],
  "overall_score": {
    "prod": "所有steps得分的乘积",
    "min": "所有steps得分的最小值",
    "last_step_score": "最后一步的得分"
  }
}

```




```
{
  "question": "原始问题描述",
  "samples": [
    {
      "steps": ["step1", "step2", "step3", ...],
      "answer": "最终答案",
      "step_values": [value1, value2, value3, ...],
      "soft_labels": [label1, label2, label3, ...]
    },
    {
      "steps": ["step1'", "step2'", "step3'", ...],
      "answer": "最终答案",
      "step_values": [value1', value2', value3', ...],
      "soft_labels": [label1', label2', label3', ...]
    },
    ...
  ]
}
```





RewardBench: Evaluating Reward Models

RewardBench: Evaluating Reward Models

Evaluating the capabilities, safety, and pitfalls of reward models

[Code](#) | [Eval. Dataset](#) | [Prior Test Sets](#) | [Results](#) | [Paper](#) | Total models: 156 | * Unverified models | ⚠ Dataset Contamination | Last restart (PST): 16:13 PST, 11 Dec 2024



⚠ Many of the top models were trained on unintentionally contaminated, AI-generated data, for more information, see this [gist](#).

🏆 RewardBench Leaderboard

🔍 RewardBench - Detailed

Prior Test Sets

About

Dataset Viewer

Model Search (delimit with ,)

☒ Seq. Classifiers

☒ DPO

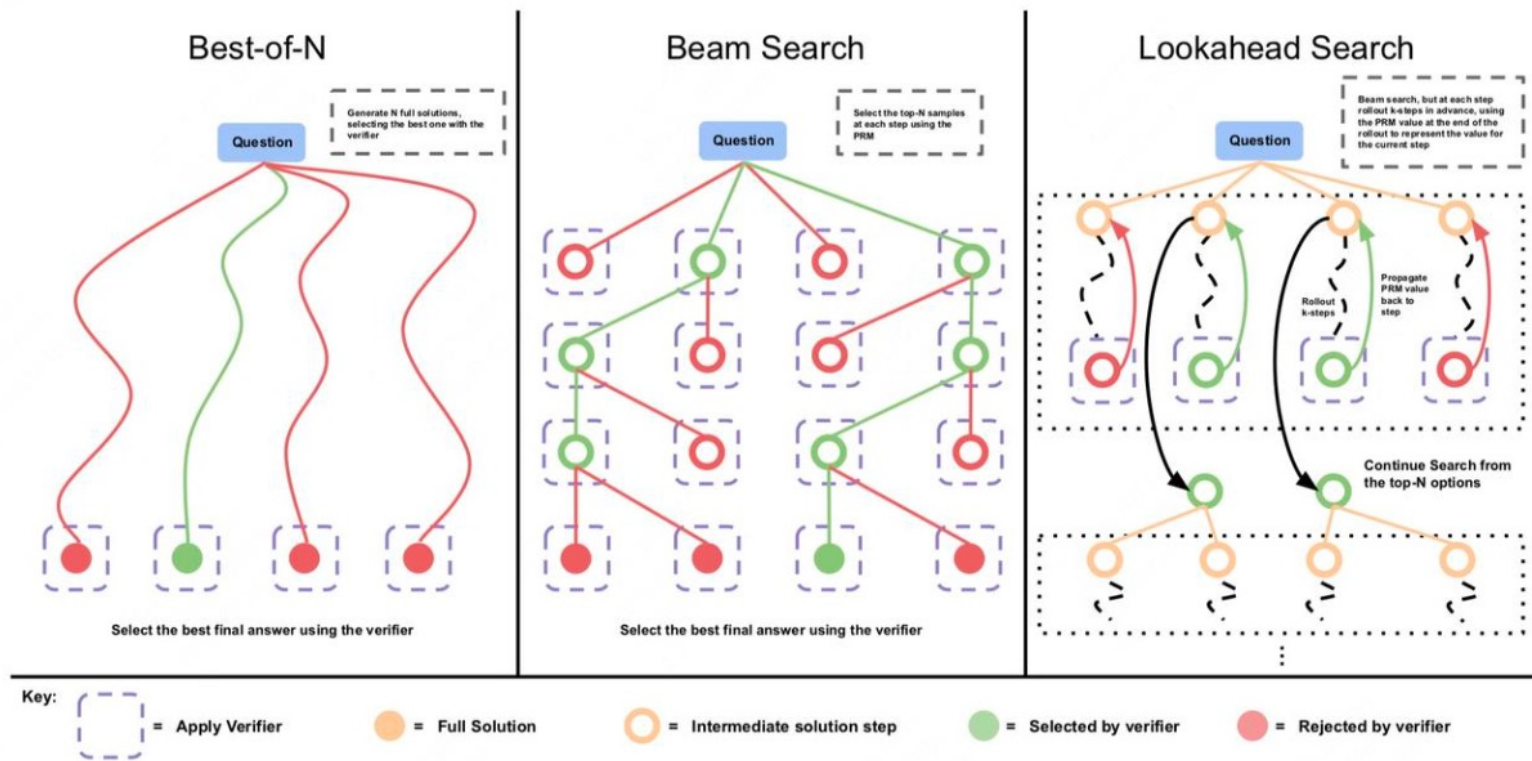
☒ Custom Classifiers

☒ Generative

☐ Prior Sets

▲	Model	▲	Model Type	▲	Score	▲	Chat	▲	Chat Hard	▲	Safety	▲	Reasoning	▲
1	infly/INF-ORM-Llama3.1-70B		Seq. Classifier		95.1		96.6		91.0		93.6		99.1	
2	Skywork/Skywork-Reward-Gemma-2-27B-v0.2		Seq. Classifier		94.3		96.1		89.9		93.0		98.1	
3	nvidia/Llama-3.1-Nemotron-70B-Reward *		Custom Classifier		94.1		97.5		85.7		95.1		98.1	
4	Skywork/Skywork-Reward-Gemma-2-27B ⚠		Seq. Classifier		93.8		95.8		91.4		91.9		96.1	
5	SF-Foundation/TextEval-Llama3.1-70B * ⚠		Generative		93.5		94.1		90.1		93.2		96.4	
6	meta-metrics/MetaMetrics-RM-v1.0		Custom Classifier		93.4		98.3		86.4		90.8		98.2	
7	Skywork/Skywork-Critic-Llama-3.1-70B ⚠		Generative		93.3		96.6		87.9		93.1		95.5	
8	Skywork/Skywork-Reward-Llama-3.1-8B-v0.2		Seq. Classifier		93.1		94.7		88.4		92.7		96.7	
9	nicolinho/ORM-Llama3.1-8B ⚠		Seq. Classifier		93.1		94.4		89.7		92.3		95.8	
10	LxzGordon/URM-LLaMa-3.1-8B ⚠		Seq. Classifier		92.9		95.5		88.2		91.1		97.0	
11	Salesforce/SFR-LLaMa-3.1-70B-Judge-r *		Generative		92.7		96.9		84.8		91.6		97.6	

使用PRM指导搜索过程



1. 采样：

- 对于一个问题，我们会采样 N 个样本。每个样本包括多个步骤（steps）和一个答案（answer）。

2. 打分：

- 调用 PRM（可能是某种预训练模型或概率模型）对这 N 个样本的中间步骤进行整体打分。
- 常用的打分方法有 prod 式、min 式和 last step 式（这些方法在之前的内容中可能已经提到过）。

3. 选择：

- 选取整体得分最高的那组步骤和答案作为输出。

1. 并行采样步骤 1（step1）

- 对于一个问题，首先并行采样 N 个 step1。例如，在上图中 N=4。这意味着在第一步中，会同时生成 4 个不同的步骤。

2. 获取 step1 的得分

- 将每个“问题 + step1”组合输入到 PRM（可能是某种预训练模型或评分模型）中，获取每个 step1 的得分。

3. 筛选出 top M 的 step1

- 从所有 step1 中选择得分最高的 M 个 step1。这一步是为了筛选出最有潜力的初始步骤。

4. 继续生成后续步骤

- 对筛选出来的 top M 的 step1 继续进行生成。每个 step1 下再生成 N 个 step2。这意味着在第二步中，每个选中的 step1 会产生 N 个新的步骤。
- 重复上述步骤来评估 step2 的分数，此时需要将“问题 + step1 + step2”组合输入到模型中进行评估。
- 这个过程会不断重复，直到达到设定的搜索停止条件为止。

1. 并行采样步骤 1（step1）

- 对于一个问题，首先并行采样 N 个 step1。例如，在上图中 N=4。这意味着在第一步中，会同时生成 4 个不同的步骤。

2. 生成后续步骤并打分

- 对于每个 step1，让它继续往下生成 K 个 step。
- 然后将最后一个 step（即生成的最后一个步骤）送入 PRM（可能是某种预训练模型或打分模型）进行打分。
- 筛选出分数最高的 top M 个结果，并返回它们对应的 step1。这些 step1 是经过筛选后的最终结果。

3. 重复步骤直至停止条件

- 从这些筛选出的 step1 出发，重复上述步骤，直到达到设定的搜索停止条件为止。



如何选择最佳搜索方法

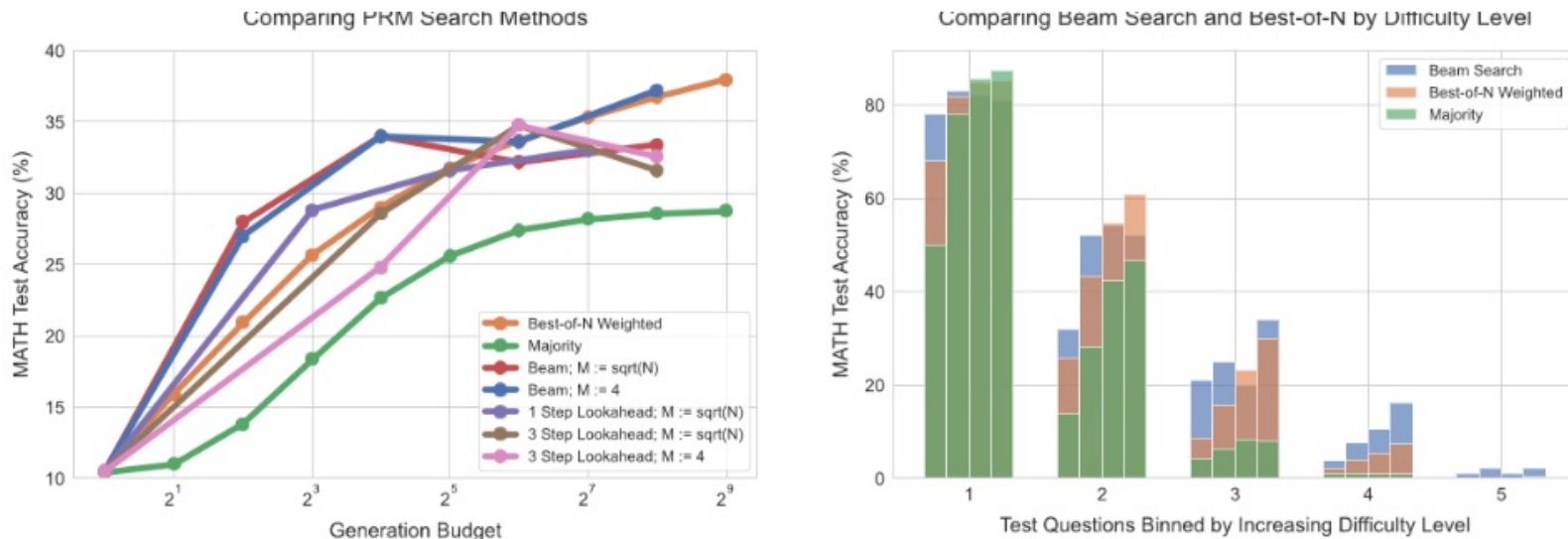


Figure 3 | Left: Comparing different methods for conducting search against PRM verifiers. We see that at low generation

• **搜索预算 (generation budget)**：这就是指采样数量 N 。即对于一个问题，你在搜索时想做多少条并行采样。我们考虑 N 的原因是GPU的算力是有限的。

- 当搜索预算较小时（即对一个问题的并行采样 N 较小时），beam search > best-of-N > lookahead
- 当搜索预算较大时（即对一个问题的并行采样 N 较大时），best-of-N > beam search > lookahead，同时在beam search的 M 超参选择合理的情况下，其效果约等于best-of-N
- 最复杂的lookahead搜索方法，表现基本处于垫底趋势。

- 当 N 较小时，在对一个问题并行采样出的 N 种结果里，可能很难直接命中最佳答案。这时通过更为复杂的搜索策略（例如beam search），从一次性采样转变为步步精心选择，可能会取得更好的结果。当 N 较大时同理可推。
- 在PRM训练得足够好的情况下，在搜索的每一步，我们可以更关注利用当前（exploit）而不是探索未来（exploration），正如前文所说，探索的本质是去寻找更好的价值函数，但它的风险是随机性较强。所以复杂的lookahead方法在我们的场景下表现并不好。

• **问题难度 (difficulty)**：对于简单的问题，模型或许利用预训练好的知识就能做回答，不需要引入复杂的搜索方法。而对于较难的问题则可能更需要设计好搜索方法引导模型步步思考。

- 在中上难度的问题上（难度3, 4），beam search > best-of-N；
- 在简单的问题上（难度1, 2），best-of-N > beam search（或者beam search略略好一些）。
- 在最难的问题上（难度5），两者的表现都不好。

在简单的问题上，模型在pretrain阶段获取的知识已能使其大概率作出正确回答。

当问题特别困难时，“PRM + 某种搜索方法”的后处理模式可能已经不能单独适用，这时或许提升模型在pretrain阶段的预训练知识更为必要。

- 搜索的效果（也即模型的推理效果）受到搜索预算和问题难度的限制，我们是在这些限制中去选择合适的搜索方法。
- 当搜索预算较小，问题较难时，更适合beam search方法，但要注意超参的调整
- 当搜索预算较大，问题较简单时，更适合best-of-N方法
- 当PRM训练得足够好时，更复杂的搜索方法（例如lookahead search）的表现可能并不好
- 当问题特别困难时，test-time scaling law的作用可能有限，这时可能有必要重新审视pretrain阶段，通过加/调整数据比例，扩张模型规模等等方式，给pretrain模型注入更多的相关知识。



改变模型的输出分布

refining the proposal distribution

**用具有高质量思考步骤的数据，直接
sft模型，能否达到同样的效果？**

那么这种方法，需要解决如何构造高质量思考步骤数据的问题



首先模型是经过sft调教过的能够按照“中间结果+答案”的形式进行结构化输出

每个正确attempts匹配上若干错误attempts, 作为一条训练数据。也就是我们的训练数据是“问题 + 若干错误attempts + 正确attempt”的形式, 这一步是让模型模拟人类思考的模式, 从步步错误的attempts中推出正确的。

If $f(x) = \frac{3x-2}{x-2}$, what is the value of $f(-2) + f(-1) + f(0)$? Express your answer as a common fraction.

Attempt 1:

```
Let's evaluate  $f(-2)$ .
We have  $f(-2) = \frac{3(-2)-2}{-2-2} = -\frac{4}{-4} = \frac{1}{1} = 1$ .
So  $f(-2) = 1$ .
Let's evaluate  $f(-1)$ .
We have  $f(-1) = \frac{3(-1)-2}{-1-2} = -\frac{5}{-3} = \frac{5}{3}$ .
So  $f(-1) = \frac{5}{3}$ .
Let's evaluate  $f(0)$ .
We have  $f(0) = \frac{3(0)-2}{0-2} = \frac{-2}{-2} = 1$ .
So  $f(0) = 1$ .
So we want to compute
 $1 + \frac{5}{3} + 1 = \frac{3}{3} + \frac{5}{3} + \frac{3}{3} = \frac{3+5+3}{3} = \frac{8}{3}$ .
So the answer is  $\boxed{\frac{8}{3}}$ .
#### \frac{8}{3}
```

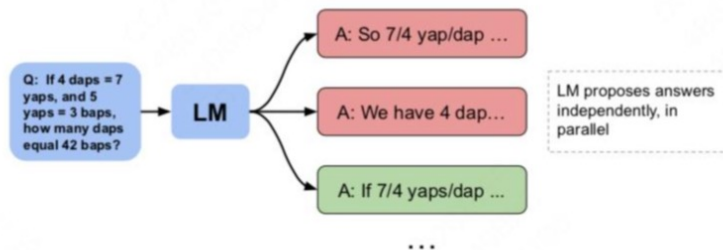
Attempt 2:

```
Let's first find  $f(-2)$ .
 $f(-2) = \frac{3(-2)-2}{-2-2} = \frac{-6-2}{-4} = \frac{-8}{-4} = 2$ .
Now let's find  $f(-1)$ .
 $f(-1) = \frac{3(-1)-2}{-1-2} = \frac{-3-2}{-3} = \frac{-5}{-3} = \frac{5}{3}$ .
Now let's find  $f(0)$ .
 $f(0) = \frac{3(0)-2}{0-2} = \frac{-2}{-2} = 1$ .
So we have  $f(-2) + f(-1) + f(0) = 2 + \frac{5}{3} + 1 = \frac{6}{3} + \frac{5}{3} + \frac{3}{3} = \frac{6+5+3}{3} = \frac{14}{3} = 5$ .
So the answer is  $\boxed{5}$ .
#### 5
```

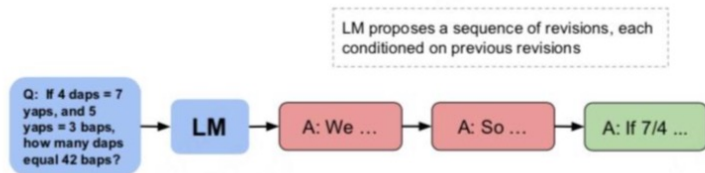
Attempt 3:

```
Let's plug in  $-2$ ,  $-1$ , and  $0$  for  $x$  in  $f(x)$ .
We get  $f(-2) = \frac{3(-2)-2}{-2-2} = \frac{-8}{-4} = \frac{2}{1} = 2$ ,  $f(-1) = \frac{3(-1)-2}{-1-2} = \frac{-5}{-3} = \frac{5}{3}$ ,  $f(0) = \frac{3(0)-2}{0-2} = \frac{-2}{-2} = 1$ .
So we have  $2 + \frac{5}{3} + 1 = \frac{6}{3} + \frac{5}{3} + \frac{3}{3} = \frac{14}{3}$ .
So the answer is  $\boxed{\frac{14}{3}}$ .
#### \frac{14}{3}
```

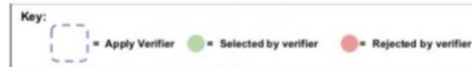
Parallel Sampling



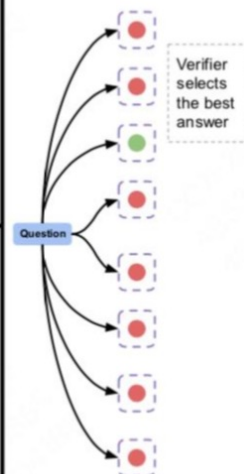
Sequential Revisions



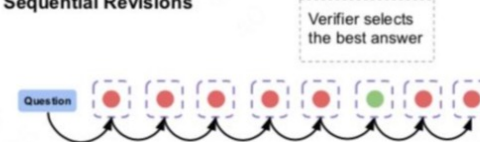
Using Revision Model + Verifier at Inference Time



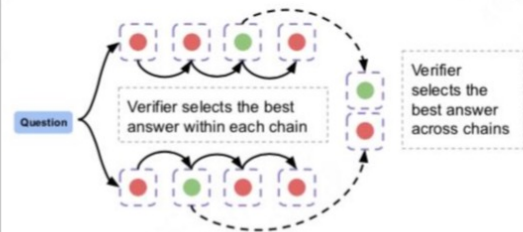
Parallel Best-of-N



Sequential Revisions



Combining Sequential / Parallel





rStar: Mutual Reasoning Makes Smaller LLMs Stronger Problem-Solvers

定义 reasoning action 构建搜索树, 5种

1. A1: propose an one-step thought. 步步推理, 每一步都有中间答案, 然后在最后一步得到答案
2. A2: propose the remaining thought steps, 一次性推理完毕, 直接得出最终答案
3. A3: propose next sub-question along with its answer. 将原始问题拆解成若干子问题并做相关回答, 最后一个字问题的答案即是最终答案
4. A4: answer the sub-question again
5. A5: rephrase the question/sub-question

根据人类思考问题的模式, 设置搜索动作空间 (Action, 缩写为A)。

搜索空间中的不同动作之间可能有前-后 (parent-child) 的依赖条件, 我们根据这些条件, 决定了一个完整的搜索树要长什么样。

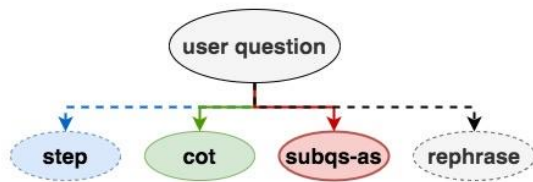
如何构造?

使用MCTS搜索最佳推理路径

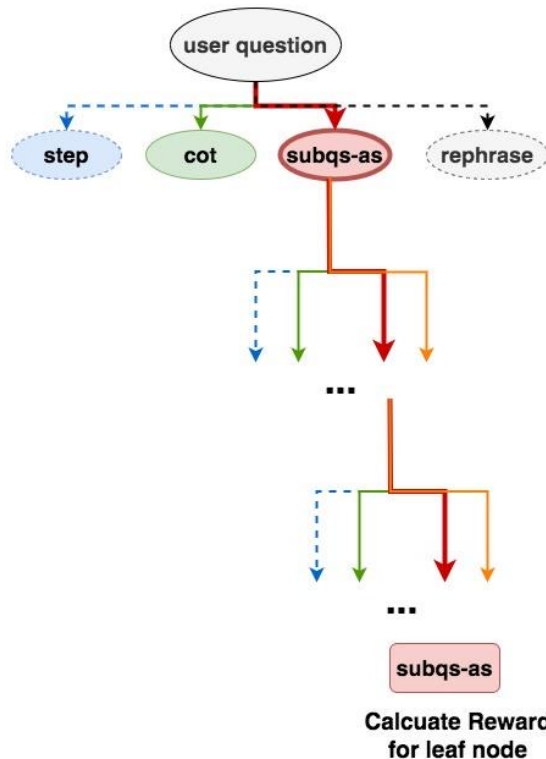
1. Select



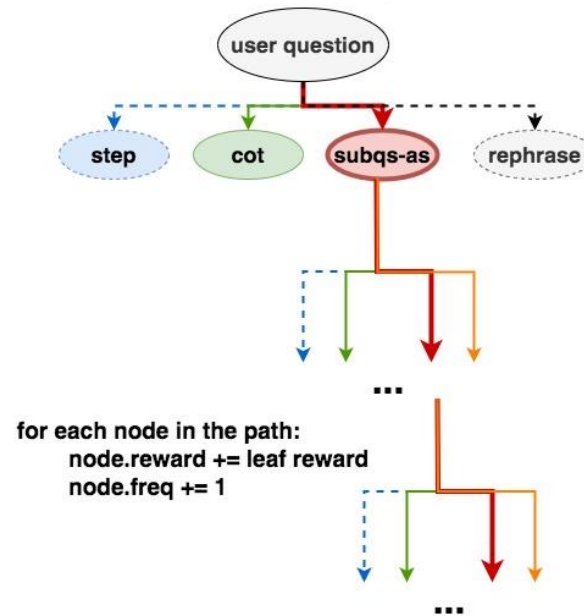
2. Expand



3. Simulate



4. backprop



A rollout = select + expand + simulate + backprop



Quiet Self-Taught Reasoner 安静的自学推理者

Quiet-STaR: Language Models Can Teach Themselves to Think Before Speaking

Eric Zelikman
Stanford University

Georges Harik
Notbad AI Inc

Yijia Shao
Stanford University

Varuna Jayasiri
Notbad AI Inc

Nick Haber
Stanford University

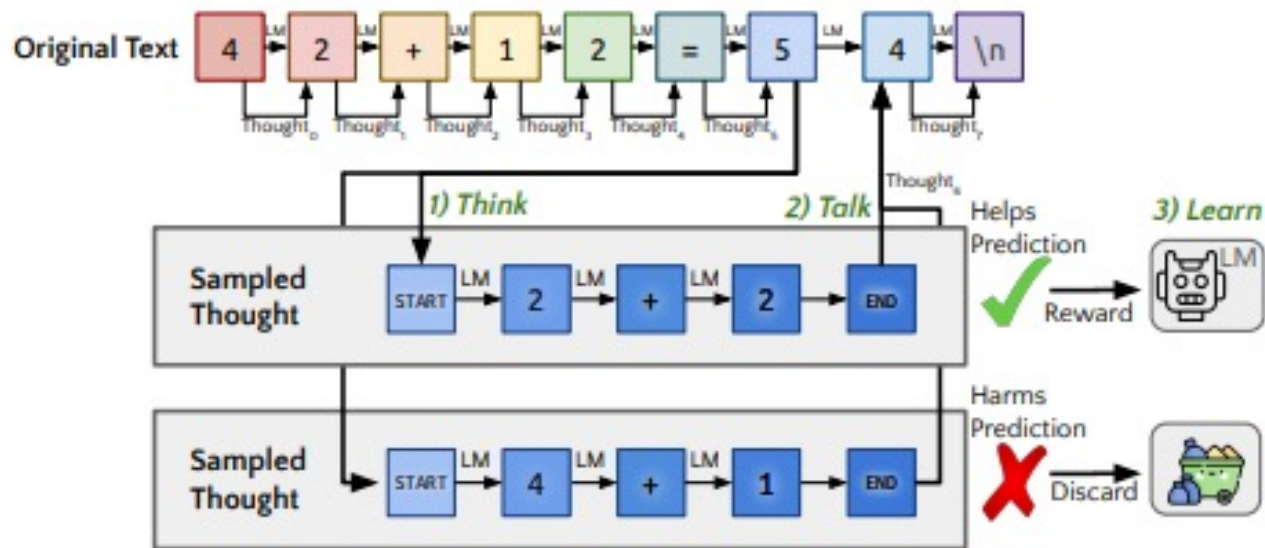
Noah D. Goodman
Stanford University

Abstract

When writing and talking, people sometimes pause to think. Although reasoning-focused works have often framed reasoning as a method of answering questions or completing agentic tasks, reasoning is implicit in almost all written text. For example, this applies to the steps not stated between the lines of a proof or to the theory of mind underlying a conversation. In the Self-Taught Reasoner (STaR, Zelikman et al. 2022), useful thinking is learned by inferring rationales from few-shot examples in question-answering and learning from those that lead to a correct answer. This is a highly constrained setting – ideally, a language model could instead learn to infer unstated rationales in arbitrary text. We present **Quiet-STaR**, a generalization of STaR in which LMs learn to generate rationales at each token to explain future text, improving their predictions. We address key challenges, including 1) the computational cost of generating continuations, 2) the fact that the LM does not initially know how to generate or use internal thoughts, and 3) the need to predict beyond individual next tokens. To resolve these, we propose a tokenwise parallel sampling algorithm, using learnable tokens indicating a thought’s start and end, and an extended teacher-forcing technique. Encouragingly, generated rationales disproportionately help model difficult-to-predict tokens and improve the LM’s ability to directly answer difficult questions. In particular, after continued pretraining of an LM on a corpus of internet text with Quiet-STaR, we find zero-shot improvements on GSM8K (5.9%→10.9%) and CommonsenseQA (36.3%→47.2%) and observe a perplexity improvement of difficult tokens in natural text. Crucially, these improvements require no fine-tuning on these tasks. Quiet-STaR marks a step towards LMs that can learn to reason in a more general and scalable way.

“Life can only be understood backwards; but it must be lived forwards.”

— Sren Kierkegaard



思考：

针对每一个输入文本序列，模型会生成多种可能的“推理过程”或“中间思路”，称为**采样思路 (Sampled Thoughts)**

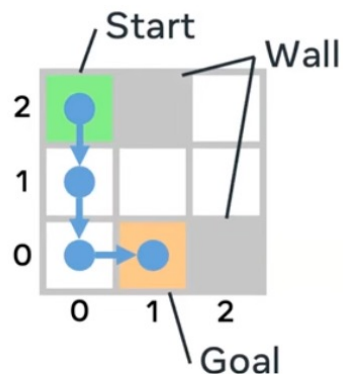
表达：

这些采样的思路被用来预测下一个目标
模型在训练时会同时利用有思路的预测路径和无思路的预测路径来进行计算

学习：

模型通过奖励机制 (如 **REINFORCE**，一种强化学习算法) 对每个“思路”的贡献进行评估

Initiating a* search as next token prediction task

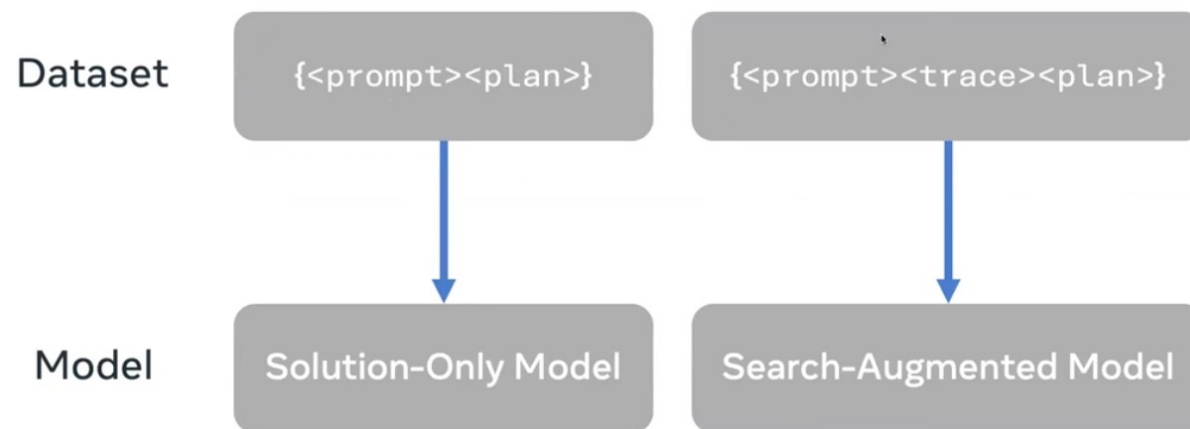


<prompt>

```
bos
start 0 2
goal 1 0
wall 1 2
wall 2 0
eos
```

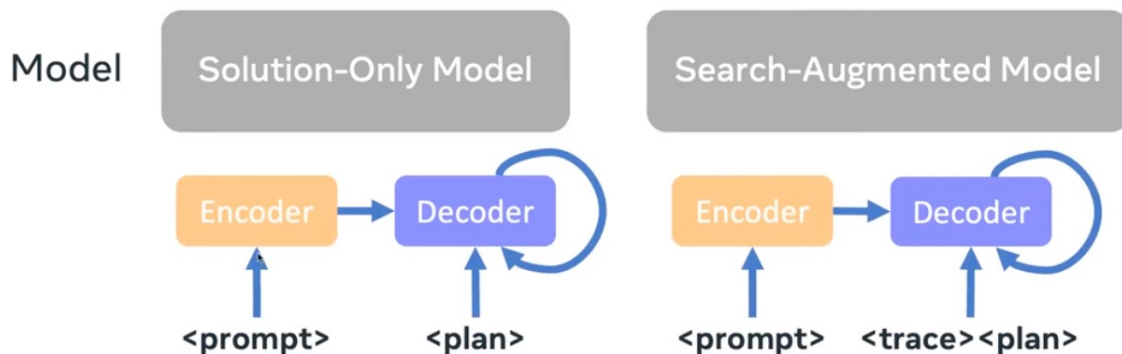
<trace><plan>

```
create 0 2 c0 c3
close 0 2 c0 c3
create 0 1 c1 c2
close 0 1 c1 c2
create 0 0 c2 c1
create 1 1 c2 c1
close 0 0 c2 c1
create 1 0 c3 c0
close 1 0 c3 c0
```

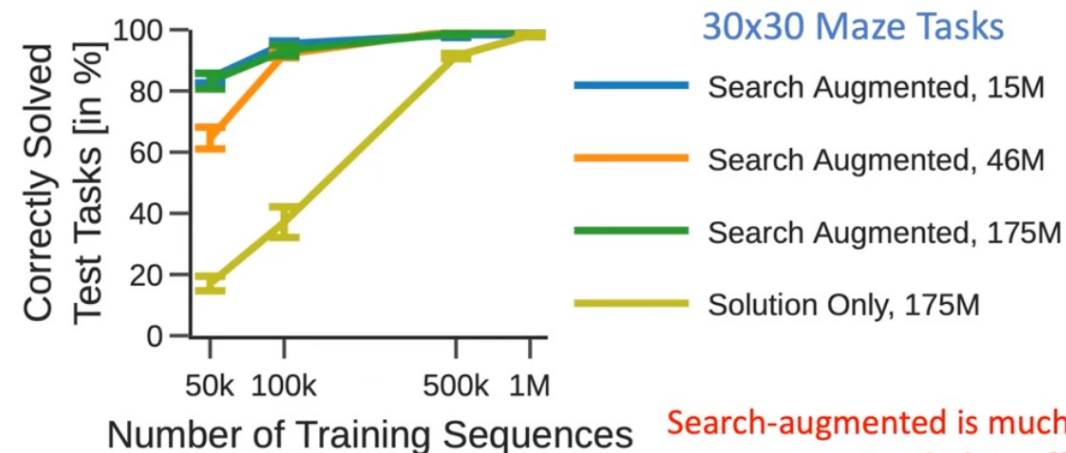


Training Method

Train a Transformer to predict the next token via teacher forcing.



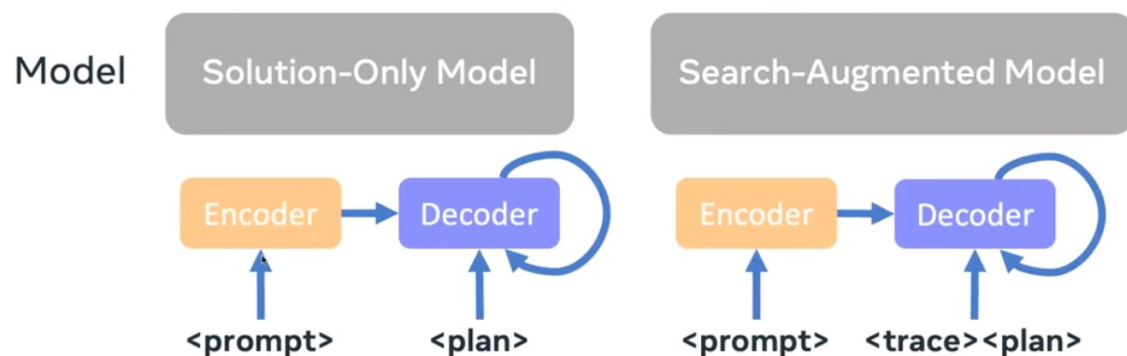
Search-Augmented vs. Solution-Only Models



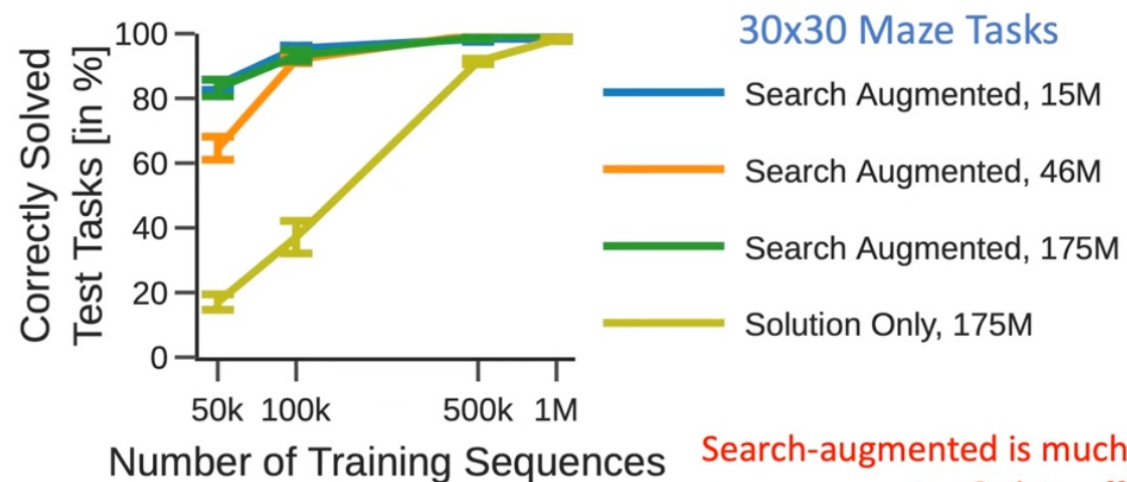
Search-augmented is much more parameter & data efficient!

Training Method

Train a Transformer to predict the next token via teacher forcing.



Search-Augmented vs. Solution-Only Models



Search-augmented is much more parameter & data efficient!



Diffusion Forcing: Next-token Prediction Meets Full-Sequence Diffusion

Boyuan Chen¹, Diego Martí Monsó^{1*}, Yilun Du¹, Max Simchowitz¹, Russ Tedrake¹, Vincent Sitzmann¹

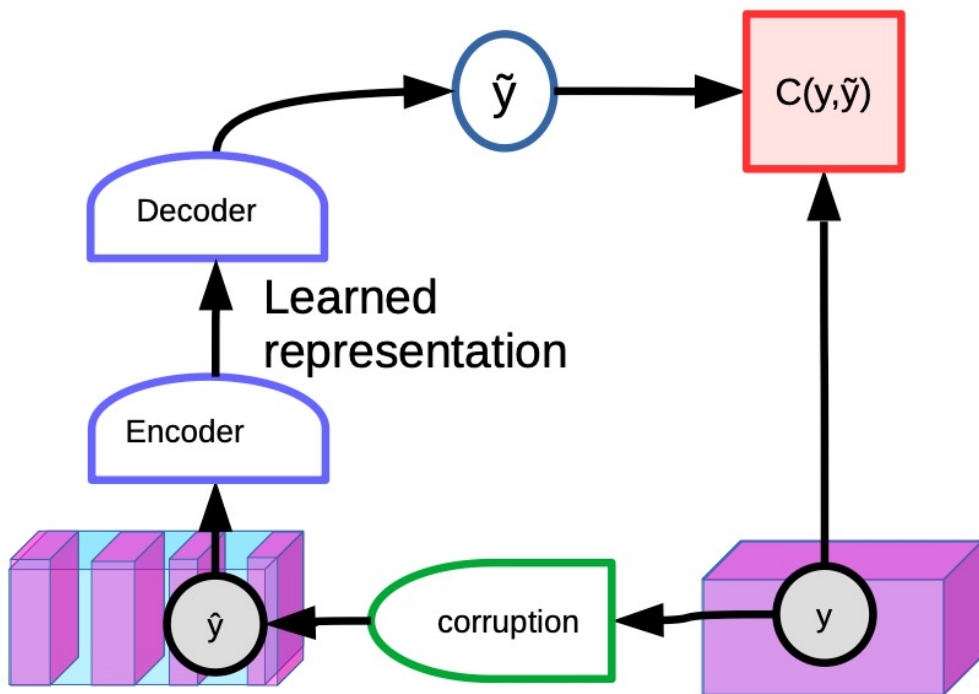
Big picture :

Self-supervised learning has taken over the world

For understanding and generating text, images, video, 3D models, speech, proteins, ...

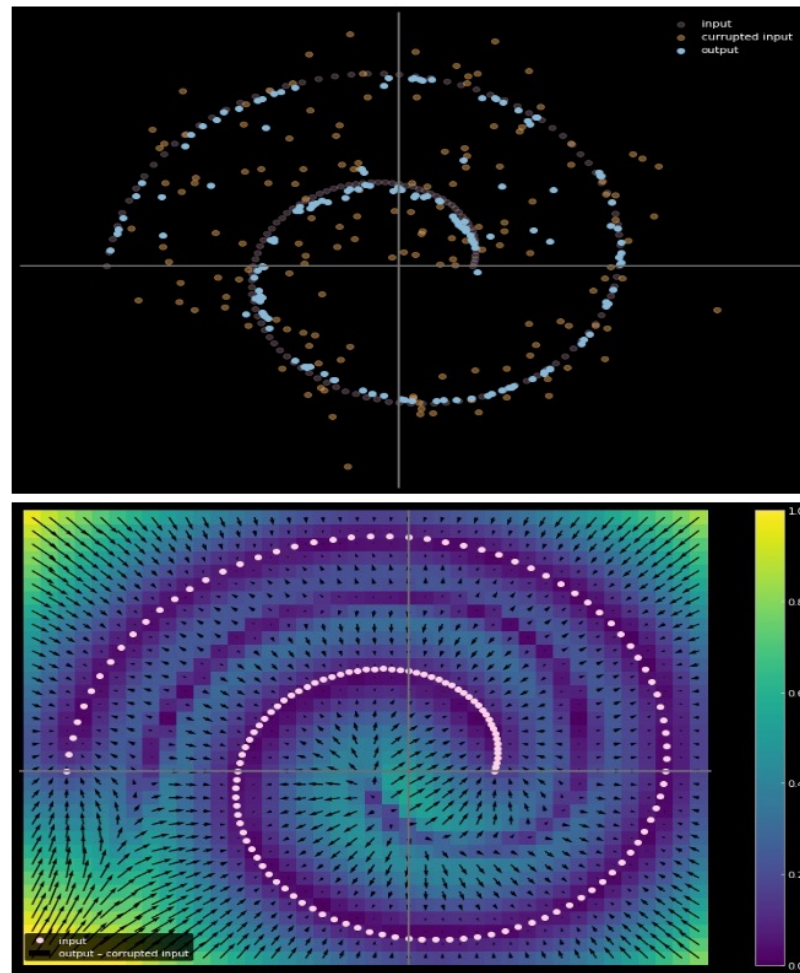
SSL via denoising / unmasking / completion

► BERT [Devlin 2018], RoBERTa [Ott 2019]



This is a [...] of text extracted
[...] a large set of [...] articles

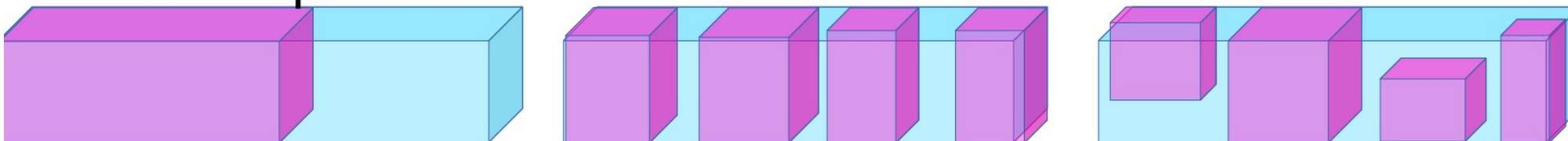
This is a piece of text extracted
from a large set of news articles



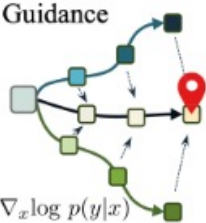
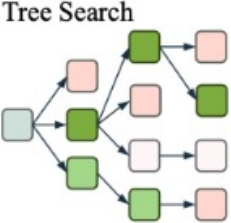
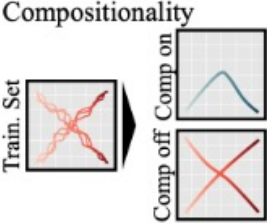
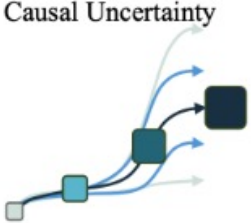
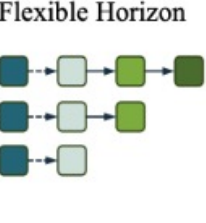
Self-supervised learning = learning to fill in the blanks

- **Reconstruct the input or Predict missing parts of the input.**

time or space →



Diffusion Forcing = “teacher forcing” + “diffusion model”

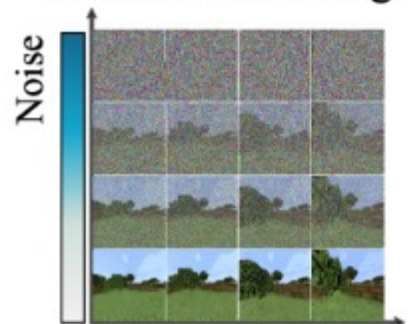
	Guidance 	Tree Search 	Compositionality 	Causal Uncertainty 	Flexible Horizon 
Teacher Forcing	✗	✓	✓	✓	✓
Full-Seq. Diffusion	✓	✗	✗	✗	✗
Diffusion Forcing	✓	✓	✓	✓	✓

Diffusion Forcing achieves so by training sequence diffusion but allowing each token to have a different noise level.

full-sequence diffusion denoise all frames at once with the same noise level,

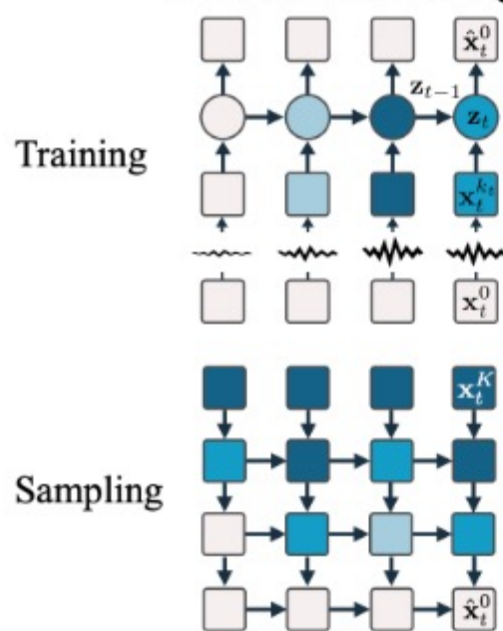
next-token prediction denoises next frame at a time with zero noise in its past tokens

Noise as Masking

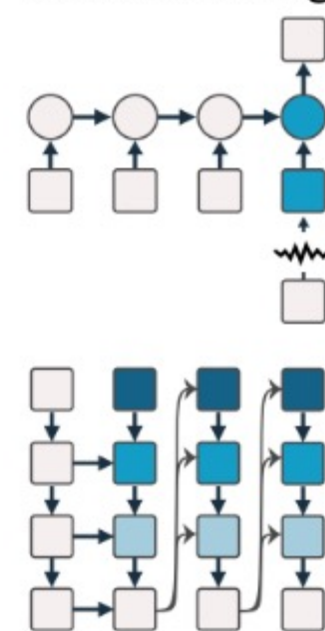


-  Observation
-  Latent State
-  Generation
-  Add Noise

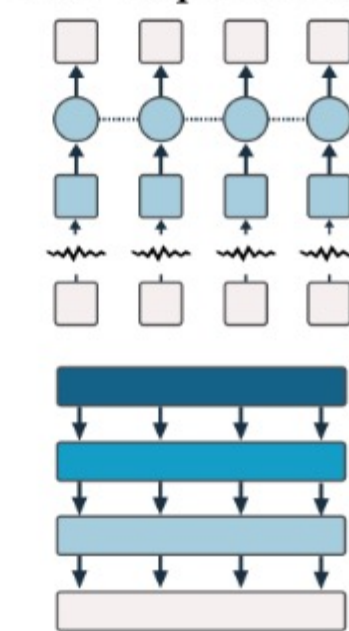
Diffusion Forcing



Teacher Forcing

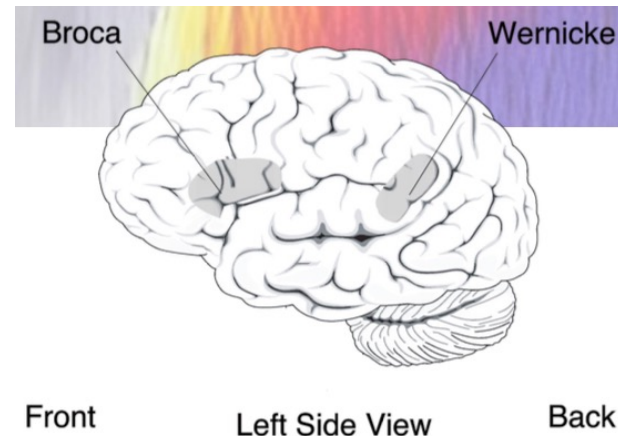


Full-Seq. Diffusion



Limitations of LLMs : no planning

- ▶ Auto-Regressive LLMs (at best) approximate the functions of the Wernicke and Broca areas in the brain.
- ▶ What about the pre-frontal cortex?



ArXiv:2301.06627

DISSOCIATING LANGUAGE AND THOUGHT IN
LARGE LANGUAGE MODELS: A COGNITIVE PERSPECTIVE

A PREPRINT

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ArXiv:2206.10498

Large Language Models Still Can't Plan
(A Benchmark for LLMs on Planning and Reasoning
about Change)

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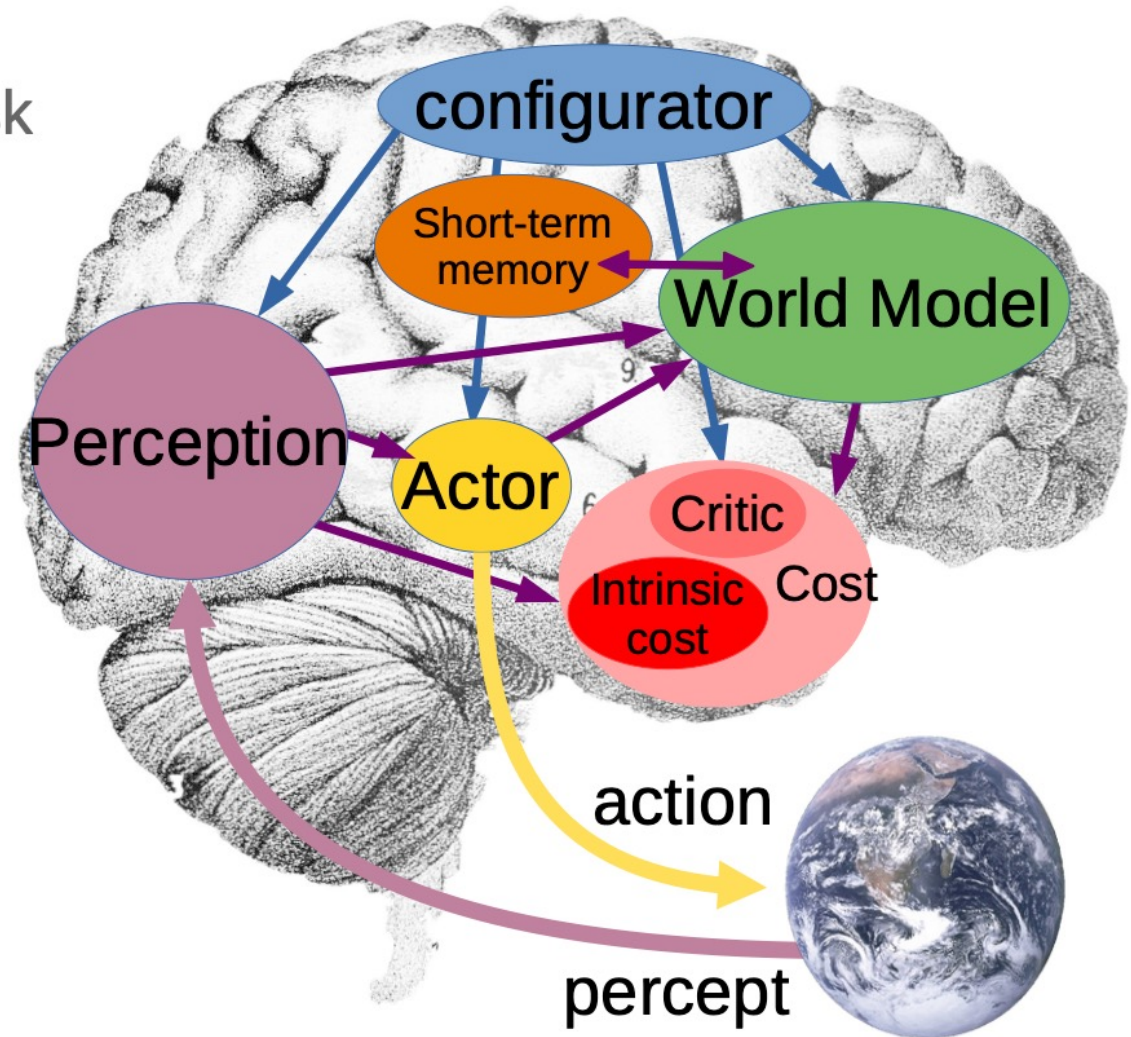
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Modular cognitive architecture for objective driven AI

- ▶ **Configurator**
 - ▶ Configures other modules for task
- ▶ **Perception**
 - ▶ Estimates state of the world
- ▶ **World Model**
 - ▶ Predicts future world states
- ▶ **Cost**
 - ▶ Compute “discomfort”
- ▶ **Actor**
 - ▶ Find optimal action sequences
- ▶ **Short-Term Memory**
 - ▶ Stores state-cost episodes



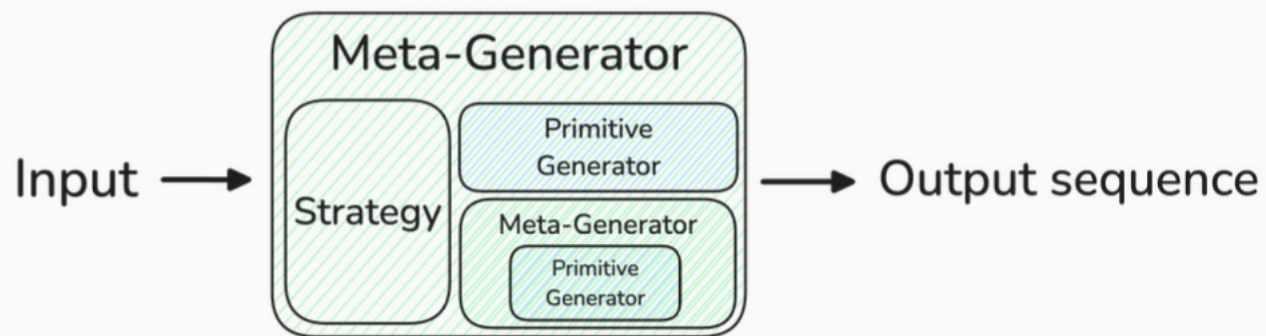


Beyond Decoding: Meta-Generation Algorithms for Large Language Models

Presenters: Matthew Finlayson, Hailey Schoelkopf, Sean Welleck

December 11, 2024

Meta-generator: High-level strategies for calling generators and using external information.



Why?

- Generate more to improve task performance
- Combine multiple models (verifiers, retrievers, ...)
- Incorporate external information (tools, feedback, ...)

From Diffusion models to Schrodinger bridges

扩散薛定谔桥

diffusion model

A diffusion process : a stochastic Markov process having continuous sample paths

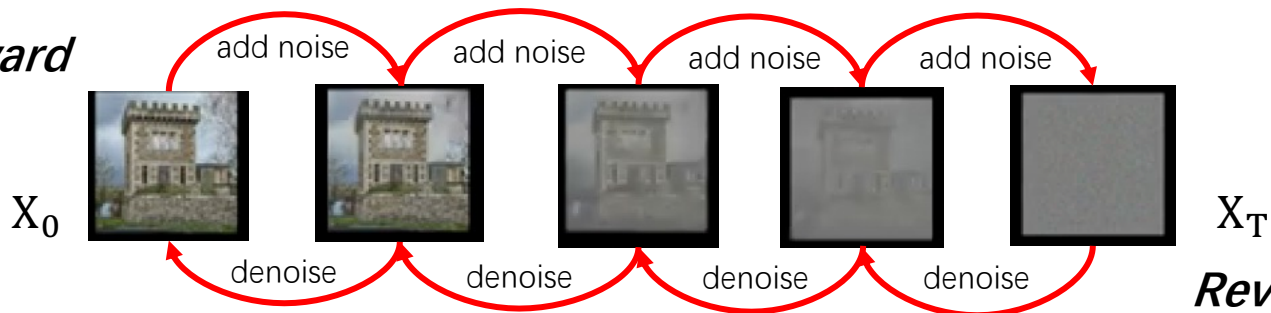
- Stochastic
- Markov process
- With no jumps

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t, \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

normal distribution → mean → variance → output

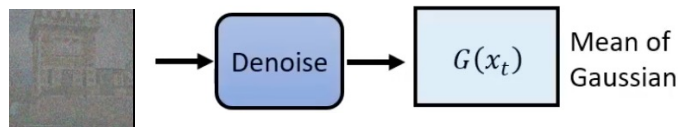
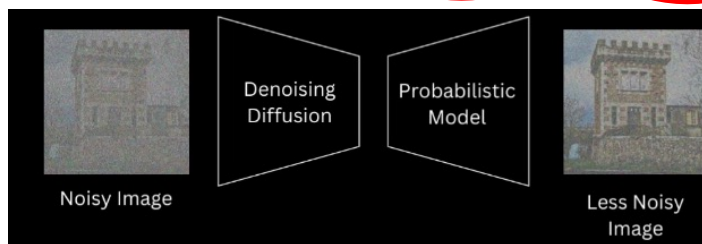
$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Forward



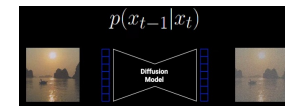
Reverse

$$P(X_t|X_{t-1}, X_{t-2}, \dots, X_0) = P(X_t|X_{t-1})$$



$$P_\theta(x_{t-1}|x_t) \propto \exp(-||G(x_t) - x_{t-1}||_2)$$

$$P_\theta(x_0) = \int_{x_1:x_T} P(x_T)P_\theta(x_{T-1}|x_T) \dots P_\theta(x_{t-1}|x_t) \dots P_\theta(x_0|x_1)dx_1:x_T$$





Diffusion model 的局限性

- 生成速度慢： Diffusion的训练过程需要分很多timestep（时间步），对连接两个分布的轨迹进行分段离散化，进而导致生成一张图片需要迭代成百上千次
- 加噪过程需要手工定义： 在加噪过程中，每个timestep加多大的高斯噪声都是手工定义的（schedule），具体什么样的schedule效果好只能通过实验尝试
- 只能单向生成： 现有的Diffusion基本都只能从高斯分布生成复杂分布（比如图像），无法在两个复杂分布之间建立映射。

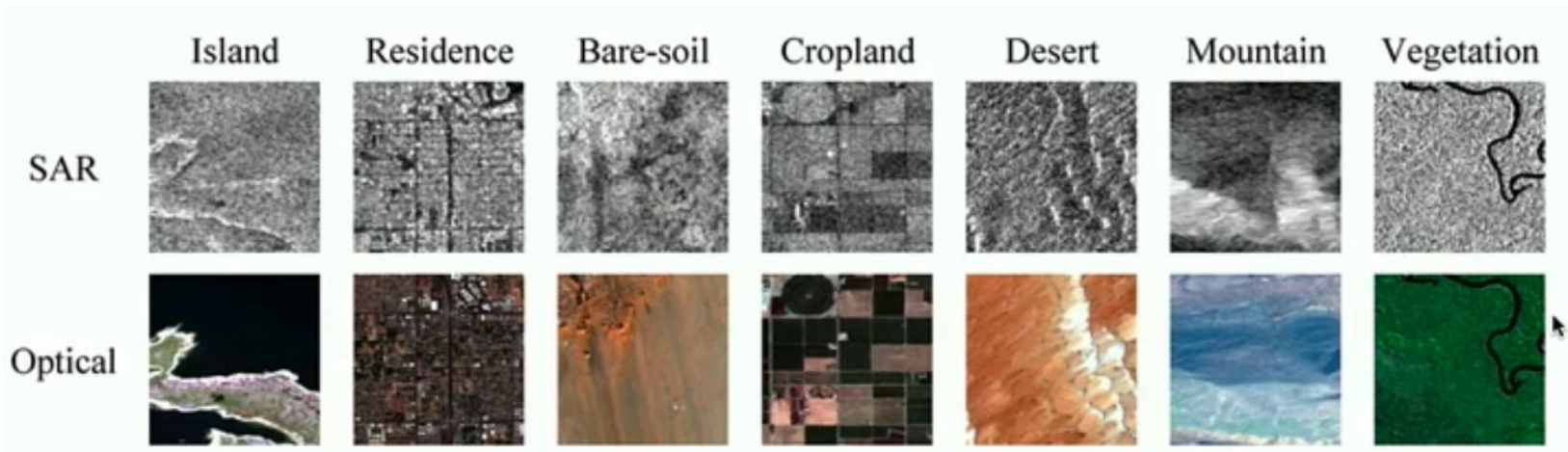
Diffusion vs DSB

- - DSB计算的是两个分布之间的最优传输（Optimal Transport, OT）；
- - DSB可以计算任意两个可采样分布之间的OT，而相比之下Diffusion只能从已知分布（如高斯噪声）开始生成数据。

Unpaired data-to-data translation



医学影像处理医学影像处理



SAR-to-optical (Wang et al, Sci. Report, 2024)



Transfer task
(Gushchin et al, ICML 2024)



薛定谔桥

P_{N+1} 为 $R^{(N+1) \times d}$ 上的概率分布所组成的空间, d 是数据维度, N 是timestamp数量

给定一个 $P_{\text{ref}} \in P_{N+1}$, 薛定谔桥是 $\pi^* \in P_{N+1}$ 满足要求:

$$\pi^* = \arg \min \{KL(\pi|p_{\text{ref}}) : \pi \in \mathcal{P}_{N+1}, \pi_0 = p_{\text{data}}, \pi_N = p_{\text{prior}}\}.$$

基本定义

- Schrödinger bridges (薛定谔桥) 是一个概率和最优传输理论相结合的概念。它源于薛定谔在 1931 - 1932 年对非平衡统计力学的思考。从数学角度看, 给定两个概率分布 μ_0 和 μ_1 (通常定义在某个状态空间, 比如 $\mathbb{R}^n$), 薛定谔桥问题是要找到一个最可能的概率测度族 $\{\mu_t\}_{t \in [0,1]}$, 使得 μ_0 和 μ_1 是这个族在 $t = 0$ 和 $t = 1$ 时的边缘分布, 并且这个测度族满足一定的随机动力学规律 (通常是马尔可夫过程)。

数学模型

- 假设系统的状态空间是 X , 在连续时间情况下, 用随机微分方程来描述系统的动态演化。设 $(\Omega, \mathcal{F}, \mathbb{P})$ 是一个概率空间, X_t 是一个随机过程, 满足 $dX_t = b(X_t, t)dt + \sigma(X_t, t)dW_t$, 其中 W_t 是维纳过程 (布朗运动), b 是漂移系数, σ 是扩散系数。

物理背景

- 在物理中, 它可以被看作是对一个物理系统在两个不同时刻的状态分布已知的情况下, 去推断系统在中间时刻状态分布的演化路径。想象有一个粒子系统, 初始时刻粒子的分布是 μ_0 , 在经过一段时间后, 最终时刻粒子的分布是 μ_1 。薛定谔桥试图找到在这个过程中粒子分布随时间最合理的变化方式, 就好像在构建一座连接初始和最终状态分布的 “桥梁”。

- 对于薛定谔桥问题, 给定初始分布 μ_0 和终端分布 μ_1 , 要找到合适的 b 和 σ 使得 $X_0 \sim \mu_0$ 和 $X_1 \sim \mu_1$, 并且在所有满足边缘分布条件的过程中, 找到使某个代价函数最小 (或某个概率测度最大) 的过程。
- 例如, 在简单的一维情况下, 设 X_t 是一个实值随机过程, 其概率密度函数 $p(x, t)$ 满足福克 - 普朗克方程 $\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}(b(x, t)p) + \frac{1}{2} \frac{\partial^2}{\partial x^2}(\sigma^2(x, t)p)$ 。薛定谔桥问题就是在给定 $p(x, 0)$ (对应 μ_0) 和 $p(x, 1)$ (对应 μ_1) 的条件下, 求解这个方程以及确定 b 和 σ 。

Generalizing Diffusion Model (广义扩散模型)

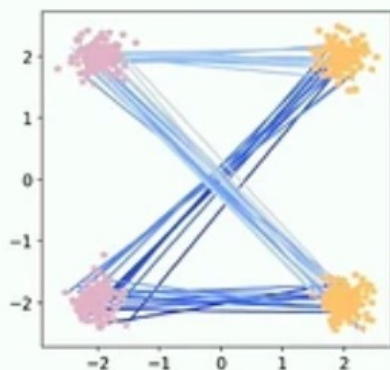
Generalizing Diffusion Models

Public

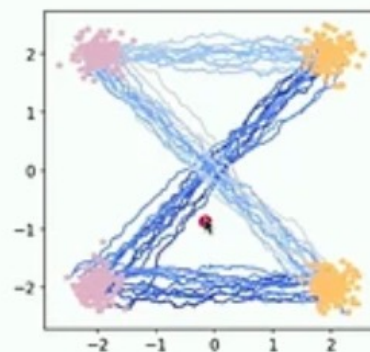
(Peluchetti 2021, Albergo et al. 2022, Liu et al. 2022)

- $X_0 \sim p_0, X_1 \sim p_1, Z \sim \mathcal{N}(0, I)$
- **Stochastic** interpolation path

$$X_t = (1 - t) X_0 + t X_1 + \sqrt{\epsilon t(1 - t)} Z \implies dX_t = \underbrace{\frac{X_1 - X_t}{1 - t} dt}_{\text{non-Markov}} + \underbrace{\sqrt{\epsilon} dB_t}_{\text{noise}}$$



$\epsilon = 0$



$\epsilon > 0$

Brownian Bridge

- samples from p_0
- samples from p_1

最优传输理论 (Optimal Transport)

- Let p_0, p_1 be fixed probability distributions.
- T is a **transport map** if $X_1 = T(X_0) \sim p_1$ for $X_0 \sim p_0$
- **Optimal Transport map** T^* minimizes expected squared Euclidean distance between X_0 and X_1

$$T^* = \operatorname{argmin}_{T \in \text{transport maps}} \mathbb{E}_{X_0 \sim p_0} [||X_0 - T(X_0)||^2]$$

- **OT** has attractive properties for unpaired data translation & generative modeling
- For empirical distributions with N samples, Kantorovich formulation cost $O(N^3)$
Entropic Optimal Transport (Cuturi, 2013) cost $O(N^2)$

Markovian Projection

- For $X_0 \sim p_0, X_1 \sim p_1$

non-Markov

$$dX_t = \frac{X_1 - X_t}{1-t} dt + \sqrt{\epsilon} dB_t$$

and

$$dX_t = \frac{\mathbb{E}[X_1 | X_t] - X_t}{1-t} dt + \sqrt{\epsilon} dB_t$$

Markov

Both ODE/SDE have **same marginals!**

$$\approx \hat{X}_{\theta^*}(t, X_t) \quad \text{Denoiser}$$

$$\theta^* = \operatorname{argmin} \mathbb{E}[\|X_1 - \hat{X}_{\theta}(t, X_t)\|^2]$$

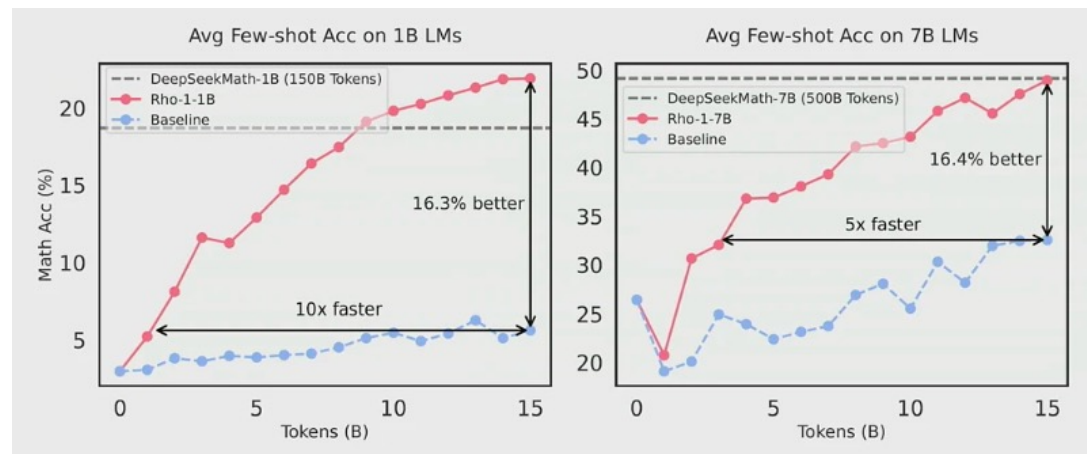
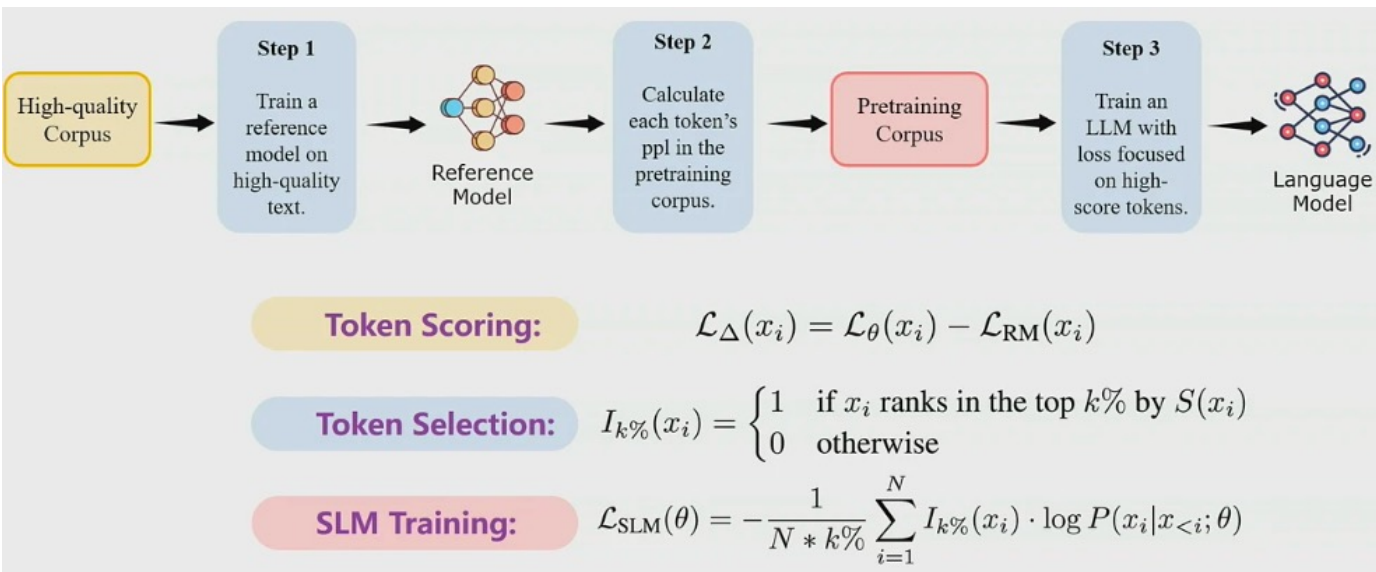
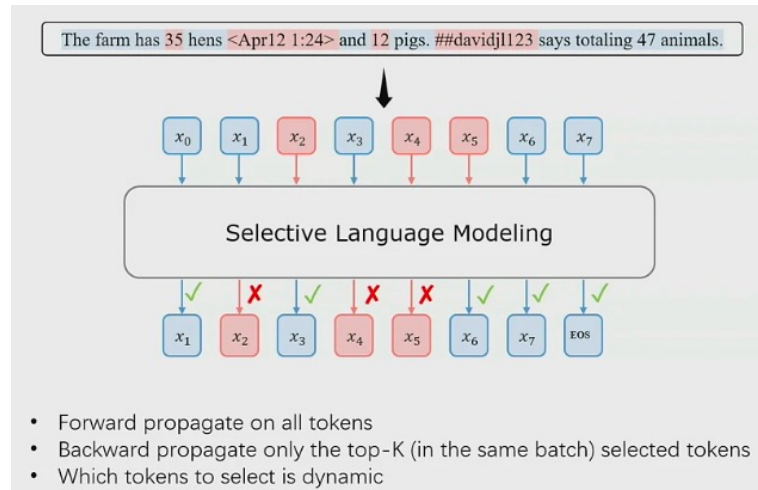
- If $\epsilon > 0$ - **Bridge Matching** (Peluchetti 2021, Albergo et al. 2022, Liu et al. 2022)
- If $\epsilon = 0$ - **Flow Matching** (Lipman et al. 2022, Liu et al. 2022, DelBracio et al. 2023)



Not All Tokens are what you need for pretraining

语料库中的所有词元对于语言模型训练并非同等重要

Rho-1 采用了选择性语言建模 (SLM)，它会选择性地针对那些与期望分布相符的有用词元进行训练。该方法先利用一个参考模型对训练词元进行评分，然后针对评分较高的词元以聚焦损失的方式来训练语言模型。

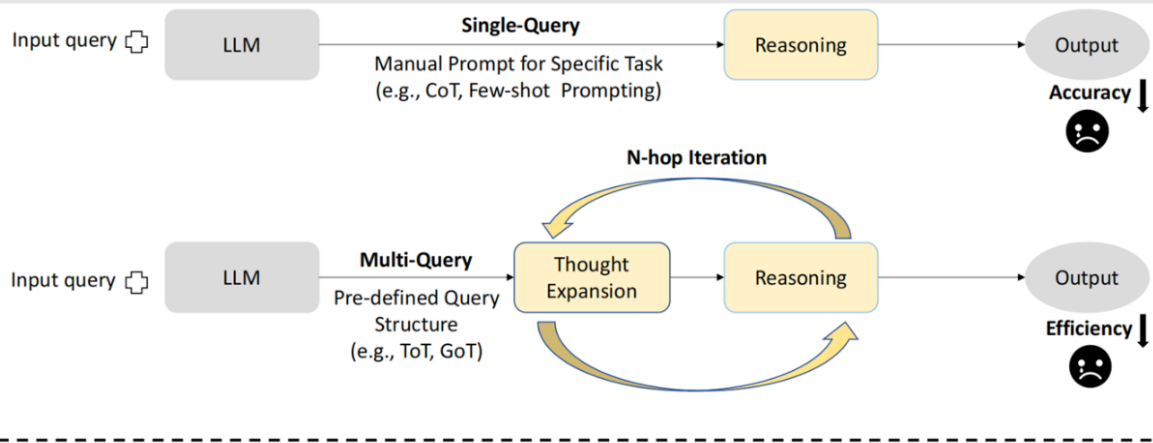




Buffer of Thoughts: Thought-Augmented Reasoning with Large Language Models

Ling Yang · Zhaochen Yu · Tianjun Zhang · Shiyi Cao · Minkai Xu · Wentao Zhang · Joseph Gonzalez · Bin CUI

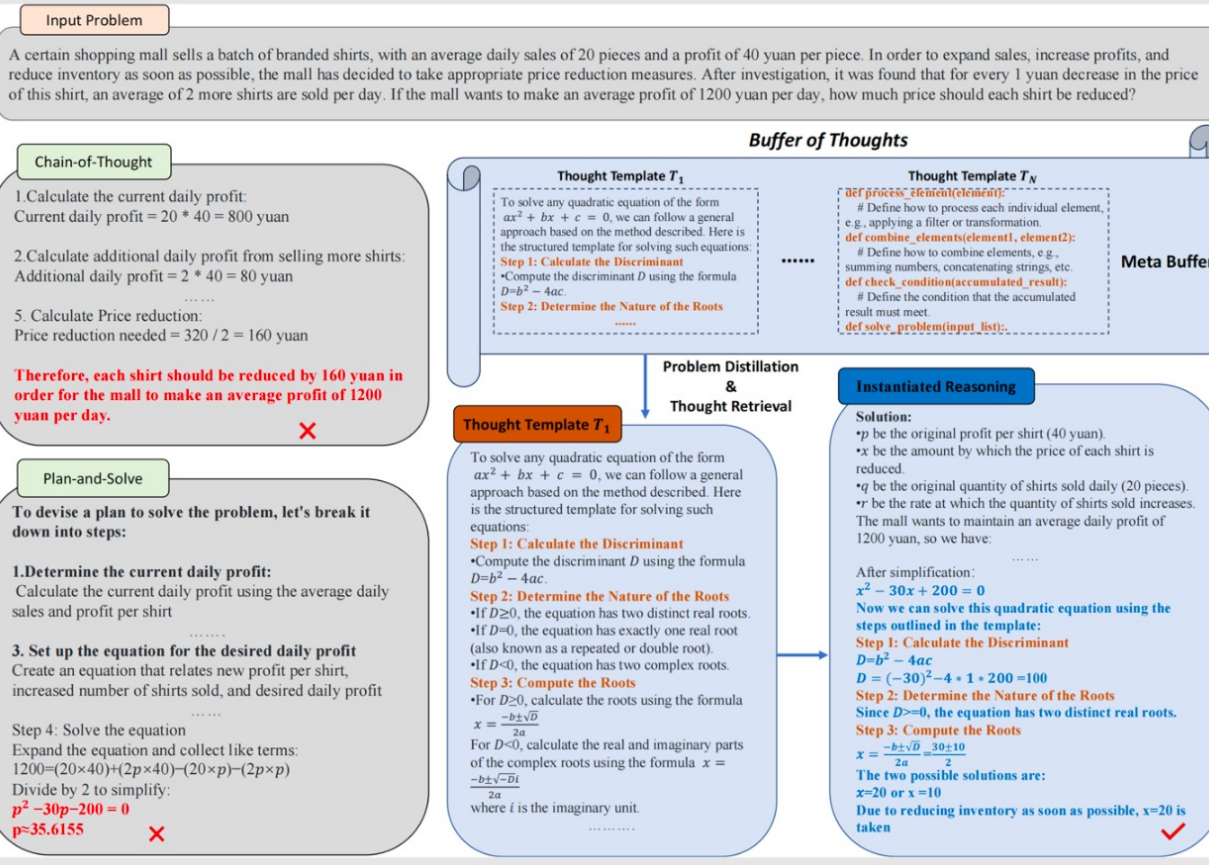
Motivation

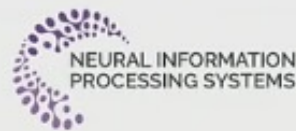


Buffer of Thoughts (BoT)

Thought-augmented Reasoning Process

Illustration of different reasoning process. Buffer of Thoughts enables large language models to tackle complex reasoning tasks through our thought-augmented reasoning process. Thought template is marked in orange and instantiated thought is marked in blue.

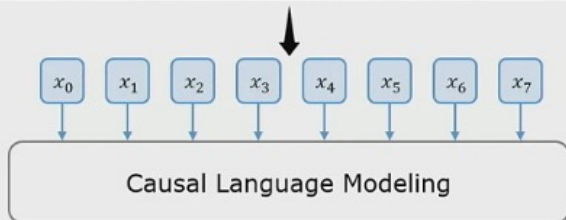




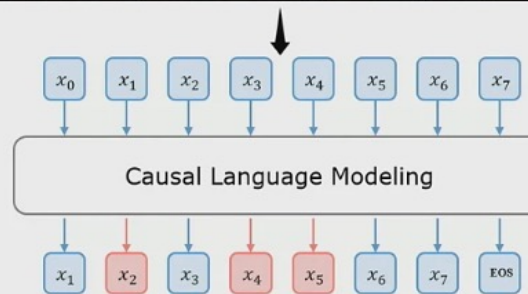
Not All Tokens Are What You Need for Pretraining

Zhenghao Lin, Zhibin Gou, Yeyun Gong, Xiao Liu, Yelong Shen, Ruochen Xu, Chen Lin, Yujiu Yang, Jian Jiao, Nan Duan, and **Weizhu Chen**

The farm has 35 hens <Apr12 1:24> and 12 pigs. ##davidjl123 says totaling 47 animals.



The farm has 35 hens <Apr12 1:24> and 12 pigs. ##davidjl123 says totaling 47 animals.



The farm has 35 hens <Apr12 1:24> and 12 pigs. ##davidjl123 says totaling 47 animals.

- Causal language model was trained left-to-right
- How human look at data:
 - zoom-in and zoom-out
 - attention to some important part (key points), ignore some unimportant parts (noise),
 - right-to-left, read-again, check consistency, ...